1.1 Real Numbers Examples

1 Distributive Property $\boxed{a}$ $2(x+3)=2x+6$ $\boxed{b}$ $(a+b)(x+y)=(a+b)x+(a+b)y$
$=ax+bx+ay+by$

2 Negatives $\boxed{a}$ $-(x+2)=-x-2$ $\boxed{b}$ $-(x+y-2)=(-x)+(-y)-(-2)=2-x-y$

3 L.C.D Add Factors $\frac{5}{36}+\frac{7}{120}=\frac{5}{2^2\cdot 3^2}+\frac{7}{2^3\cdot 3\cdot 5}=\frac{5(2\cdot 5)}{36\cdot(2\cdot 5)}+\frac{7\cdot 3}{120\cdot 3}=\frac{50+21}{360}=\frac{71}{360}$

4 Sets Union & Intersection $S\cup T=\{1,2,3,4,5,6,7\}$, $S\cap T=\{4,5\}$, $S\cap V=\emptyset$

5 Graphing Intervals $[-1,2]=\{x \mid -1\le x<2\}$, $[1.5,4]=\{x \mid 1.5\le x\le 4\}$, $(-3,\infty)=\{x \mid -3<x\}$

6 Finding Unions & Intersections $(1,3)\cap[2,7]=[2,3)$, $(1,3)\cup[2,7]=(1,7]$

7 Absolute Values abcd $|3-\pi|=-(3-\pi)=\pi-3$

8 Distance $|-8-2|=|-10|=10$